READ
FULL

Backup of my 'University Practice Sheets'

This should be divided into sub-sections based on file type. Things i want to be downloaded directly? Links. General stuff with low amt of code-lines? Copy-paste right in here! (Anyways I will still put fillers for my amnesia-prone future self)

CLI Cmds

- 04/08/2026

```
magick clipboard: cli.png
##first use imagemagick dwld and then use snipping tool
```

Note: This is supposed to be a very useful gadget for Windows users. Idk i mean Mac already saves it unless you drag it someewhere.

- 28/07/2026

```
wd -d .
#go to the designated folder in GUI and type this on the window to directly access Powershell
cmd
#same way, cmd also words and cmd is ultimate destination even from Powershell
copy /filepath .
#copies to this folder without creation of new name


```

> Note: Powershell is somewhat important for Windows. cmd from Powershell.

- 21/07/2026

```
# Win + R -> cmd
dir
#Listings special: cd . -> Current folder, cd . . -> Parent folder
#3rd fn of cd -> Shows name of current folder
cd | clip
#piping, copying the folder name to clipboard
start .
#opens folder in which you are in
/?
#help
#no touch in Windows, directly echo smth to get the file out of nowhere
>
#overwrite
>>
#append
type file_name
#views file in cmd
copy file_name1 file_name2
#even when second file DNE
copy ..\a\*.txt ..\c
#everything in one line when you are in folder b
/p
#pauses output aka asks y/n every step e.g., deletion
/s
#forces, force del like when you have smth in the dir you wanna rm
```

> Note: This is real work from that day. Yeah life is difficult at first when you work on UNIX and university works in Windows. Also, comment on next line for the perspective colouration to kick in.

Python

- 13/10/2026

```
#exercise 5
from turtle import *
mode('logo')
rt(30)
for i in range(3):
    fd(50); rt(120)

done()

#exercise6
import random
from turtle import *

L = [0, 90, 180, 270]
L2 = ['red', 'blue', 'green', 'purple']

reset()
mode('logo')

for i in range(20):
    color(random.choice(L2))
    rt(random.choice(L))
    fd(30)

done()

#exercise8

n = int(input('Enter a number to get its multiplication table:'))
for i in range(1,101):
    print(n, 'x', i, '=', n*i)

#exercise10

from turtle import *
reset()
speed(0)
color('gold')
pensize(10)

for i in range(10):
    forward(250)
    right(144)

done()

#Exercise2
#11
st = "gcatgactattccttgac"
ac = st.count('a')
cc = st.count('c')
gc_ = st.count('g')
tc = st.count('t')

print('The counts of a,c,g and t are respectively: ', ac, ',', cc, ',', gc_, ',', tc)

#12
st = "gcatgactattccttgac"
print(st[::-1])

#13
```

> 3
> 7
> 3
> 8
> 3
> 9
> 4
> 0
> 4
> 1
> 4
> 2
> 4
> 3
> 4
> 4
> 4
> 5
> 4
> 6
> 4
> 7
> 4
> 8
> 4
> 9
> 5
> 0
> 5
> 1
> 5
> 2
> 5
> 3
> 5
> 4
> 5
> 5
> 5
> 6
> 5
> 7
> 5
> 8
> 5
> 9
> 6
> 0
> 6
> 1

- 01/10/2026

> `eval` must get your strings in quotes for it to know its string otherwise there will be error.
> Review IPython (and Jupyter Notebook) cmds. `%` kind.
> What is `sys.argv`?

- 29/09/2026

```
from turtle import *
dir()
from os import system
system('cls')
-- can be given in reverse order too --
mode('logo') #opens turtle screen
reset()
---
fd(100); rt(90); fd(100); rt(90); fd(100); rt(90); fd(100); rt(90) #turtle draws square
#do same thing with while loop
>>> from time import sleep
>>> n_i = 0
>>> while n_i < MaxSteps:
    ...      fd(100); sleep(5); rt(90)
    ...      n_i += 1

-- tmod.py --
from turtle import *
mode('logo')
reset()
def drwsqr(l, c):
    color(c)
    for _ in range(4):
        fd(l); rt(90)
```

The coordinate system is entirely based on turtle's current position. By def in python the turtle pointer faces right. `reset()` puts the screen back to canvas. You can execute redundant steps faster using `for` loop. If you write the code elsewhere for example a file named `tmod.py` then you must do `import tmod`. So then you treat it like a module and pulling fns from it would look like `tmod.fn()`

- 25/08/2026

```
>>> dir( ) #gives all the built-in functions in whatever I am in
>>> help(atan2)
Help on built-in function atan2 in module math:

atan2(y, x, /)
    Return the arc tangent (measured in radians) of y/x.

    Unlike atan(y/x), the signs of both x and y are considered.
#help function for any help from the documentations

#While in python shell, one can run commands of the system too using os and sys
>>> os.system('cmd of system')
0

# Function defintion, include useful parts with multiline comments
>>> def cls():
...     '''
...     cls: clears the terminal
...     '''
...     os.system('cls')
...
>>> help(cls)
Help on function cls in module __main__:

cls()
    cls: clears the terminal
```

Note: Function definition also done in class. Download 'automate the boring stuff with python' after going home. In python, unlike basic prompt keeping pressing the up arrow does get you prev cmds. `^D` used for ending prompt.

Markdown

- 04/08/2026

```
## Command Line Interface and DOS Experience
### Creating a work folder
### Creating text files
### Using echo and re-direction
```

Note: Put this in hw lab notebook. Put directory of your mac too and a whole section on your experience on a mac which is gonna be different.

- 28/07/2026

```
**Apple**
#Bold
*Apple*
#Italics
#We are using marktext for md files
#In MarkText Ctrl+P to go to Editor View.

![ ](imgName.png ) #insertion of images
# ! means 'show me'

Hey this is an eqn $ inline eqn $

$$ in new para line to initiate a lone standing eqn then type your eqn:
```

> Note: This *could* have been taught on rentry but ... nvm ig. MarkText is a markup text renderer. Work has been put in Garden.

Plotting

- 29/09/2026

```
gnuplot> set zeroaxis
gnuplot> plot 'data.txt'
gnuplot> plot 'data.txt'
gnuplot> plot 'data.txt' u 1:2, 'data.txt' u 1:3

---
# you could take diff smoothing techniques for diff data also you can change using ' . ' to using ' - . -'
too.
#This is a drooping exponential curve
Least fitting squares (?)
reset
f(x) = a*exp(-b*x)
a = 300
b = 1
fit f(x) 'data.txt' u 1:2 via a,b
plot 'data.txt' w p pt 1 ps 4, 'data.txt' smooth csplines, f(x) lw 3

#and these are ig the final vals
print a, b
415.727591880944 0.725452619243909

#further
plot 'data.txt' index 0 w p pt 1 ps 4, 'data.txt' index 1 w p pt 4 ps 3
```

and `fit.log` looks like:

```
*******************************************************************************
Tue Sep 29 12:36:11 2026


FIT:    data read from 'data.txt' u 1:2
        format = x:z
        #datapoints = 5
        residuals are weighted equally (unit weight)

function used for fitting: f(x)
    f(x) = a*exp(-b*x)
fitted parameters initialized with current variable values

iter      chisq       delta/lim  lambda   a             b
   0 1.2956971770e+04   0.00e+00  5.95e+01    3.000000e+02   1.000000e+00
   5 3.6694325224e+01  -8.05e-02  5.95e-04    4.157276e+02   7.254526e-01

After 5 iterations the fit converged.
final sum of squares of residuals : 36.6943
rel. change during last iteration : -8.0484e-07

degrees of freedom    (FIT_NDF)                        : 3
rms of residuals      (FIT_STDFIT) = sqrt(WSSR/ndf)    : 3.49735
variance of residuals (reduced chisquare) = WSSR/ndf   : 12.2314

Final set of parameters            Asymptotic Standard Error
=======================            ==========================
a               = 415.728          +/- 14.76        (3.551%)
b               = 0.725453         +/- 0.02464      (3.397%)

correlation matrix of the fit parameters:
                a      b
a               1.000
b               0.904  1.000
```

using a list (do a tab b/w vals or two spaces at least)

'#' x	y1	y2
1.2	2.3	2.5
2.4	4.5	4.3
3.2	5.6	5.7
4.5	7.6	9.5

- 18/08/2026

```
gnuplot> $data << EOD
1.56 0.9998
EOD #inserting data into a block
gnuplot> plot h(x) lw 2, '$data' with points ps 4

gnuplot> load 'mysine_plt.txt' #loading code from external source

#setting titles and labels
gnuplot> plot [-2*pi:2*pi] f(x) lw 2 title 'sine curve', g(x) lw 2 title 'cosine curve'
gnuplot> set xlabel 'x'
gnuplot> set ylabel 'y'

 #you can shift line w/o breaking with '\'

gnuplot> #reset to put back default settings. It is good practice to put that in beginning of script
gnuplot> show sample

        sampling rate is 100, 100

gnuplot> plot sin(1/x)
gnuplot> set samples 1000
gnuplot> replot
```

- 11/08/2026

```
f(x) = sin(x)
plot f(x)
#plotted smth
set zeroaxis
plot [-pi:pi] f(x) ##specifying range
plot [-2*pi:2*pi] [:] f(x) ls 2 lw 3
plot [-pi:pi] [:] sin(x) ls 2 lw 3, cos(x) ls 7 lw 3 lc 'blue' #def opt is line for graph
```

Files

> *Things in this world are volatile. They disappear unless you take a snapshot ^-^*

Table of PDFs

Date:	File:	Remarks:

Link to my github:

Lab Reports

Edit Export

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